Introduction to Causal Inference with Hidden Confounders

Instruments and Proxies

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Observation and a treatment decision

Observed comparison

$$\mathbb{E}[Y \mid T = 1] - \mathbb{E}[Y \mid T = 0]$$

How do outcomes differ between the groups we observe?

Intervention

$$\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$$

What would change if we assigned treatment differently?

The lecture plan

Part I. Foundations — 35 minutes

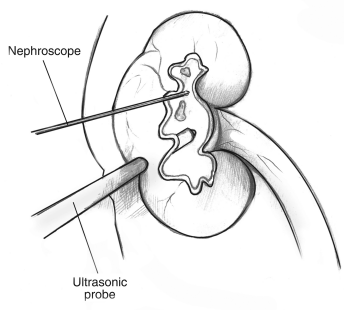
Confounding, potential outcomes, randomization and adjustment.

Part II. Hidden confounders — 50 minutes

Instruments, then proxy variables and bridge functions.

Comparison and practical handoff — 5 minutes

Kidney stones: the overall comparison



B: percutaneous nephrolithotomy

	Successful / treated	Rate
A	$273/350$	78.0%
B	$289/350$	82.6%

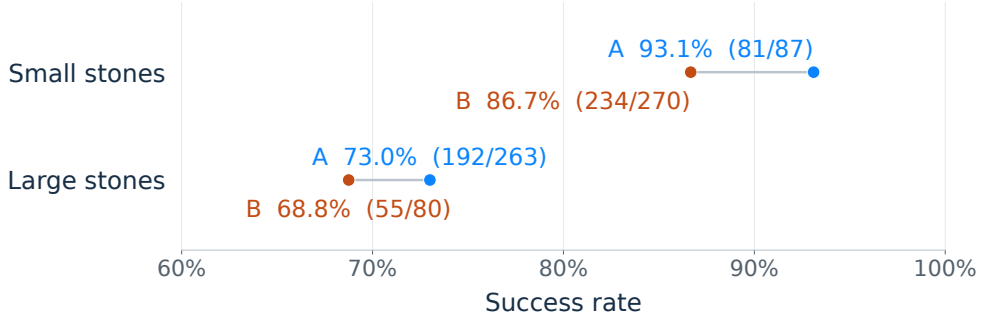
A is open surgery.

$Y = 1$: treatment success at three months.

What else could differ between these cohorts?

Counts: Charig et al. (1986). Illustration: NIDDK, National Institutes of Health, media asset 17731.

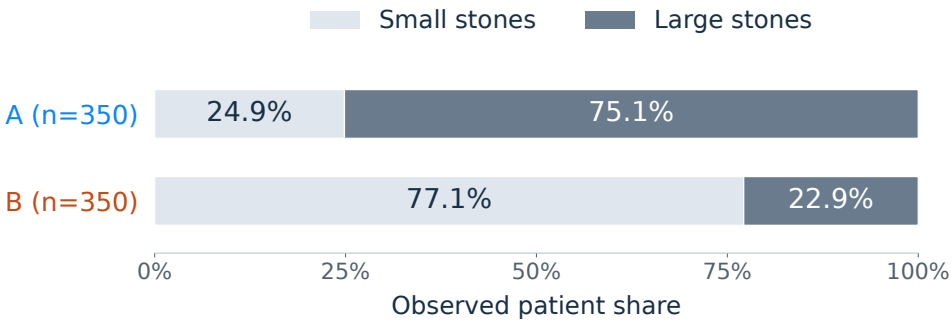
The comparison reverses within groups



A has higher success in both stone-size groups.

Charig et al. (1986). Exact counts appear beside the rates; no sampling-uncertainty claim is made.

Different patient mixtures



A treated more large-stone cases. Which mix should a fair comparison use?

Patient counts: Charig et al. (1986). Bar widths are proportional to the observed case mix.

Sunscreen and sunburn

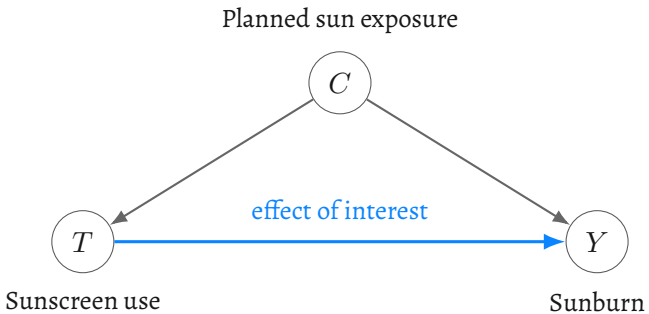
Suppose sunscreen users have more sunburn in an observational dataset.

A possible explanation

People planning more sun exposure may be more likely to use sunscreen.

Does that comparison tell us what assigning sunscreen would do?

The causal path and the common cause



The observed association can combine both paths.

Hypothetical causal graph. Arrows encode assumptions; the treatment effect remains in the graph.

One unit, two potential outcomes

Unit	Treatment	Observed	Missing
i	1	$Y_i(1)$	$Y_i(0)$
j	0	$Y_j(0)$	$Y_j(1)$

$$\text{Individual effect} = Y_i(1) - Y_i(0).$$

Each unit reveals only one potential outcome.

Neyman (1923); Rubin (1974).

A well-defined treatment comparison

Treatment versions
Specify the intervention and the outcome measurement time.

No interference
Another unit's treatment does not change $Y_i(t)$.

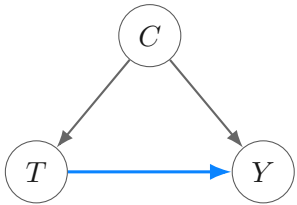
Consistency

$$Y_i = Y_i(T_i) = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

Hernán and Robins (2020), Chapters 1–3.

Selecting a group and assigning treatment

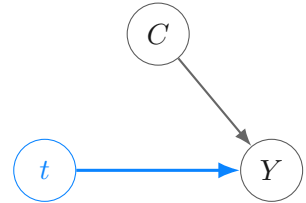
Observe $T = t$



$$\mathbb{E}[Y \mid T = t]$$

Select those who received t.

Set $T = t$



$$\mathbb{E}[Y \mid \mathrm{do}(T = t)]$$

Assign t in the target population.

Pearl (2009). Setting treatment replaces its assignment mechanism.

Target, identification and estimation

Target

$$\mu(t) = \mathbb{E}[Y(t)], \qquad \tau = \mu(1) - \mu(0).$$

Identification

Assumptions connect $\mu(t)$ to the observed distribution.

Estimation

A finite sample approximates that observed-data formula.

What randomization supplies

$$T \perp (Y(0), Y(1)), \qquad \Pr(T = 1) = p \in (0, 1).$$

$$\begin{aligned} \mu(t) &= \mathbb{E}[Y(t) \mid T = t] && \text{randomization} \\ &= \mathbb{E}[Y \mid T = t] && \text{consistency.} \end{aligned}$$

Randomization identifies population means. Individual counterfactuals remain missing.

Hernán and Robins (2020), Chapters 2–3.

When treatment is chosen

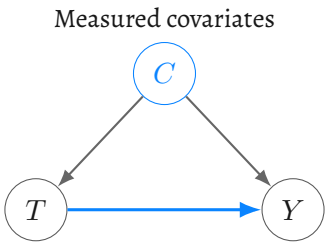
Patients, clinicians or decision systems may choose treatment using causes of the outcome.

The adjustment idea

Compare treatments within suitable measured groups C.

Then average over one common population.

The adjustment assumptions



C must be a sufficient adjustment set.

Exchangeability

$$Y(t) \perp T \mid C.$$

Positivity

$$\Pr(T = t \mid C = c) > 0.$$

Maintain consistency.

Hernán and Robins (2020), Chapters 3 and 7.

The adjustment derivation

$$
\begin{aligned}
\mu(t) &= \mathbb{E}_C\{\mathbb{E}[Y(t) \mid C]\} && \text{iterated expectation} \\
&= \mathbb{E}_C\{\mathbb{E}[Y(t) \mid T = t, C]\} && \text{exchangeability} \\
&= \mathbb{E}_C\{\mathbb{E}[Y \mid T = t, C]\} && \text{consistency.}
\end{aligned}
$$

Compare within C, then use the population weights $\Pr(C)$.

Returning to kidney stones

Use 51% small and 49% large cases for **both** treatments.

$$\widehat{\mu}(A) = .51\frac{81}{87} + .49\frac{192}{263} = 83.25\%,$$

$$\widehat{\mu}(B) = .51\frac{234}{270} + .49\frac{55}{80} = 77.89\%.$$

A minus B changes from -4.57 to $+5.37$ percentage points.

Causal interpretation still requires size to be a sufficient adjustment set.

Charig et al. (1986). Historical observational cohorts; possible calendar-time confounding remains.

Estimating the adjusted mean

Outcome regression

$$\widehat{\mu}(t) = \frac{1}{n} \sum_i \widehat{m}_t(C_i).$$

Fit the response in each group, then average predictions for everyone.

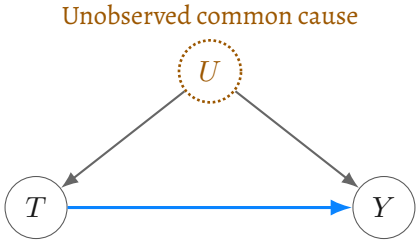
Other implementations

Weight by the propensity, or combine both nuisance models.

All rely on the same causal assumptions.

Hernán and Robins (2020), Chapters 12–13. Weighting and augmented formulas: appendix A2.

A required confounder is hidden



$$\mu(t) = \mathbb{E}_U\{\mathbb{E}[Y \mid T = t, U]\}.$$

This adjustment calculation cannot be computed from (T, Y) alone.

The same-data/different-effect construction appears in appendix A10 and the practical.

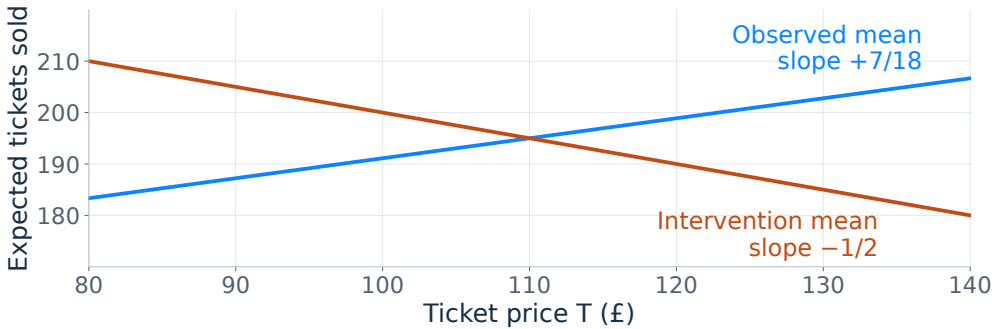
Part II: hidden confounders

An instrument
Additional treatment variation with justified exclusion and exogeneity.

Two proxies
Additional measurements of hidden confounding, with different restrictions.

The target remains an effect of changing treatment.

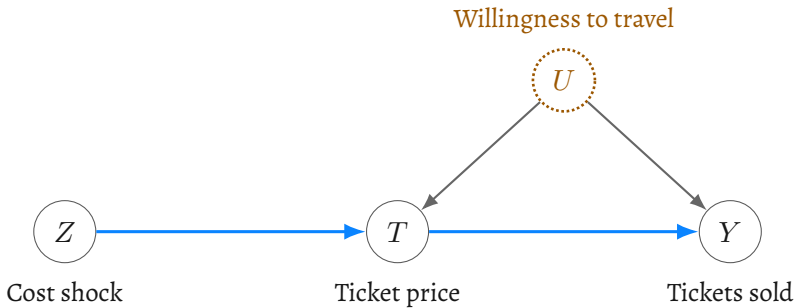
Price and sales can move together



The same constructed model has a positive observed slope and a negative causal slope.

Constructed population, exact conditional/intervention means. T: price in pounds; Y: tickets sold.

A candidate instrument



The proposed cost shock changes sales only through price.

Common shocks or changes in flight capacity could invalidate this story.

The model behind the IV formula

$$Y(t) = \alpha + \beta t + \varepsilon, \qquad Y = Y(T).$$

β is a common causal slope. T may depend on ε.

Exclusion No outcome effect of Z beyond T.

Exogeneity $\mathbb{E}[\varepsilon \mid Z] = 0$.

Relevance $\mathrm{Cov}(Z, T) \neq 0$.

Imbens (2014). The derivation identifies the common slope specified here.

The IV covariance calculation

$$\begin{aligned}\operatorname{Cov}(Z, Y) &= \beta \operatorname{Cov}(Z, T) + \operatorname{Cov}(Z, \varepsilon) \\ &= \beta \operatorname{Cov}(Z, T) \qquad \text{by exogeneity.}\end{aligned}$$

$$\beta = \frac{\operatorname{Cov}(Z, Y)}{\operatorname{Cov}(Z, T)}$$

Relevance makes the denominator nonzero.

The binary Wald ratio

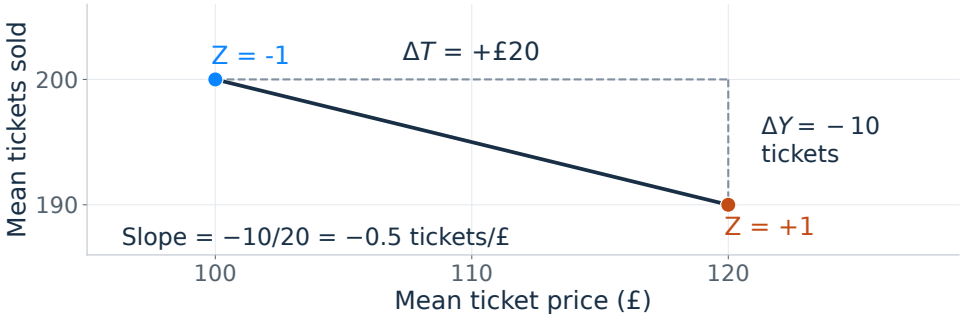
Cost-shock level	Mean price	Mean tickets sold
$Z = 0$	£100	200
$Z = 1$	£120	190

$$\beta = \frac{190 - 200}{120 - 100} = -0.5 \quad \text{tickets per pound.}$$

The causal interpretation uses the preceding IV model and restrictions.

Illustrative binary-instrument population. A causal answer requires validity and a specified effect model.

Instrument-induced mean movements



Across instrument means, $-10/20 = -0.5$ tickets per pound.

Constructed continuous-Z population; conditional levels $Z = -1, +1$. Exact means; generator in appendix A16.

Two-stage least squares

1. Fit treatment using the instruments

$$\widehat{\mathbf{t}} = P_{\mathbf{Z}}\mathbf{t}.$$

2. Regress the outcome on fitted treatment

$$\widehat{\beta}_{\text{2SLS}} = \frac{\widehat{\mathbf{t}}^{\top}\mathbf{y}}{\widehat{\mathbf{t}}^{\top}\widehat{\mathbf{t}}}.$$

This is a sample linear projection. Use IV standard errors.

Imbens (2014). Full sample matrix formula and rank conditions: appendix A13.

IV: validity, strength and target

Validity

Can the exclusion and exogeneity restrictions be justified?

Strength

Does the instrument move treatment enough for stable estimation?

Target

The model here identifies a common slope.

What if we have measurements of U, but no credible instrument?

Imbens (2014); Angrist, Imbens and Rubin (1996). Heterogeneous-effect extension: appendix A3.

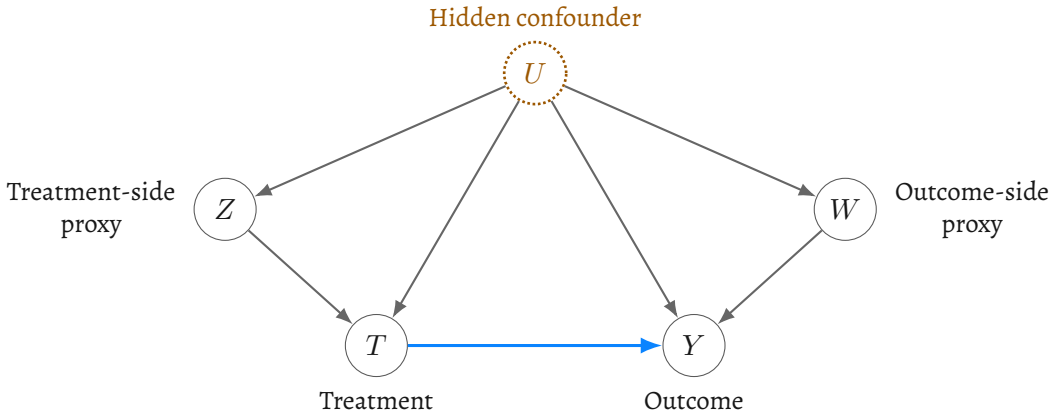
Two proxy roles

U	hidden common causes of treatment and outcome
$Z = M_T$	indexes observed groups in which the bridge must match outcome means
$W = M_Y$	supplies the variable transformed by the bridge and averaged

Here Z is a proxy, with a different role from the preceding instrument.

Miao et al. (2018); Tchetgen Tchetgen et al. (2020).

The proxy causal structure



U remains hidden. We observe (Z, W, T, Y).

Miao et al. (2018). Dotted nodes are unobserved; missing paths encode restrictions.

The proxy restrictions

Treatment-side proxy

$$Y \perp Z \mid (T, U).$$

Given T, U, Z adds no outcome information.

Outcome-side proxy

$$W \perp (T, Z) \mid U.$$

Given U, W is independent of the joint pair (T, Z).

Miao et al. (2018); Tchetgen Tchetgen et al. (2020).

Adjustment if U were observed

For binary T, assume consistency and, for each t,

$$Y(t) \perp T \mid U, \qquad \Pr(T = t \mid U) > 0.$$

$$a_t(U) = \mathbb{E}[Y \mid T = t, U], \qquad \mu(t) = \mathbb{E}_U[a_t(U)].$$

We need an average that can be computed without observing U.

The useful latent bridge relation

Suppose an integrable h_t satisfies

$$a_t(U) = \mathbb{E}[h_t(W) \mid U].$$

Then iterated expectation would give

$$\mu(t) = \mathbb{E}_U[a_t(U)] = \mathbb{E}_W[h_t(W)].$$

The final average uses W, but this proposed relation still contains U.

Teaching progression: Gretton, MBZUAI 2026 Part 2, slides 25–27. Explicit sufficient conditions follow.

An equation using observed variables

The proxy restrictions turn the proposed latent bridge into

$$\mathbb{E}[Y \mid Z, T = t] = \mathbb{E}[h_t(W) \mid Z, T = t].$$

Z	indexes the groups whose means must match
h_t	transforms W into outcome units

Solving this equation still needs a causal justification.

Observed bridge equation. Forward projection alone is not the causal identification proof.

The missing causal condition

Maintain the causal and proxy restrictions.

Existence
An integrable observed bridge h_t solves the equation.

Latent completeness
For each t and every $g \in L^1(P_U)$,

$$\mathbb{E}[g(U) \mid Z, T = t] = 0 \text{ a.s.} \quad \Longrightarrow \quad g(U) = 0 \text{ a.s.}$$

Require $\mathbb{E}|Y(t)| < \infty$ and $\mathbb{E}|h_t(W)| < \infty$.

Miao et al. (2018); Tchetgen Tchetgen et al. (2020). Full sufficient statement: appendix A14.

Why the bridge average is causal

Write $r_t(U) = \mathbb{E}[h_t(W) \mid U]$.

$$\mathbb{E}[a_t(U) - r_t(U) \mid Z, T = t] = 0 \qquad \text{bridge and restrictions}$$

$$a_t(U) = r_t(U) \qquad \text{latent completeness.}$$

$$\mu(t) = \mathbb{E}[a_t(U)] = \mathbb{E}[r_t(U)] = \mathbb{E}[h_t(W)] \; .$$

Causal assumptions, then completeness, then iterated expectation. Expanded proof: appendix A4.

Discrete proxies give a linear system

$$\underbrace{K_t}_{\text{mixtures}} \quad \underbrace{h_t}_{\text{bridge}} = \underbrace{m_t}_{\text{means}}.$$

$$K_t[z, w] = \Pr(W = w \mid Z = z, T = t), \qquad m_t[z] = \mathbb{E}[Y \mid Z = z, T = t].$$

Rows index Z. Columns index W.

A square nonsingular K_t gives $h_t = K_t^{-1} m_t$.

Miao et al. (2018). Matrix rank concerns solving the observed equation.

A fully specified binary example

$$U \sim \text{Bernoulli}(1/2), \qquad Y(t) = 2t + 4U.$$

	$U = 0$	$U = 1$
$\Pr(T = 1 \mid U)$	$1/4$	$3/4$
$\Pr(Z = U \mid U)$	$3/4$	$3/4$
$\Pr(W = U \mid U)$	$3/4$	$3/4$

Generate T, Z, W independently conditional on U.

Constructed population. Exact enumeration and both treatment-specific systems: appendix A17.

Two equations determine the bridge

At $t = 0$, the two Z groups require

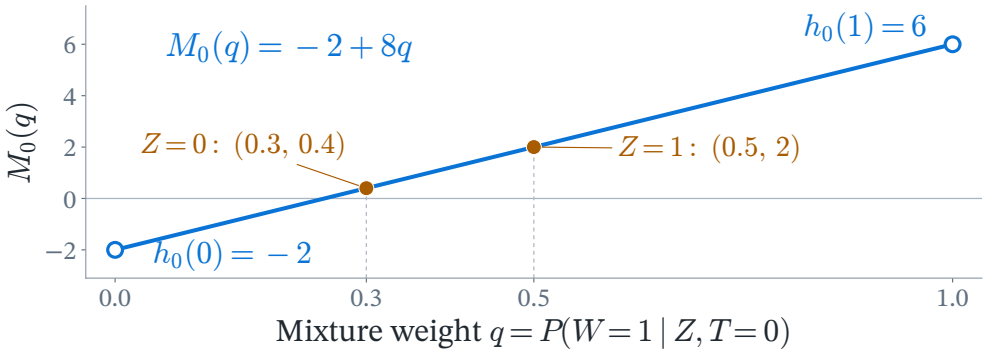
$$0.7h_0(0) + 0.3h_0(1) = 0.4,$$

$$0.5h_0(0) + 0.5h_0(1) = 2.$$

$$h_0(0) = -2, \qquad h_0(1) = 6.$$

Exact observed moments from the preceding binary model.

The bridge calibrates the two mixtures



One pair of bridge values matches both observed group means.

Exact binary example. The horizontal coordinate is a mixture weight; W has only the values 0 and 1.

One population average for each treatment

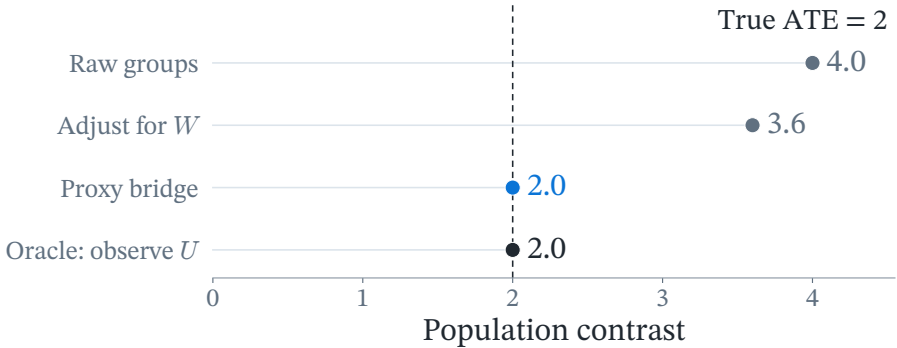
	$W = 0$	$W = 1$	Population average
Population weights	$1/2$	$1/2$	
Bridge h_0	-2	6	2
Bridge h_1	0	8	4

$$\mu(t) = p_W^\top h_t, \qquad \tau = 4 - 2 = 2\,.$$

Use the marginal population weights p_W.

Exact bridge values and marginal weights in the constructed model.

A proxy can leave confounding



In this model, directly adjusting for W leaves confounding.

Constructed binary population, exact enumeration. Oracle U is available only to the simulator.

A sum becomes an integral

Discrete W

$$m_t(z) = \sum_w h_t(w) \Pr(W = w \mid Z = z, T = t).$$

Continuous W

$$m_t(z) = \int h_t(w) p(w \mid Z = z, T = t)\, dw.$$

We now solve for a function. This is a Fredholm equation of the first kind.

Miao et al. (2018). Function spaces, existence and inverse stability: appendix A9.

The estimation calculation

1. Estimate the moments

$\widehat{K}_t, \widehat{m}_t$ and population weights $\widehat{p}_W$.

2. Fit a regularized bridge

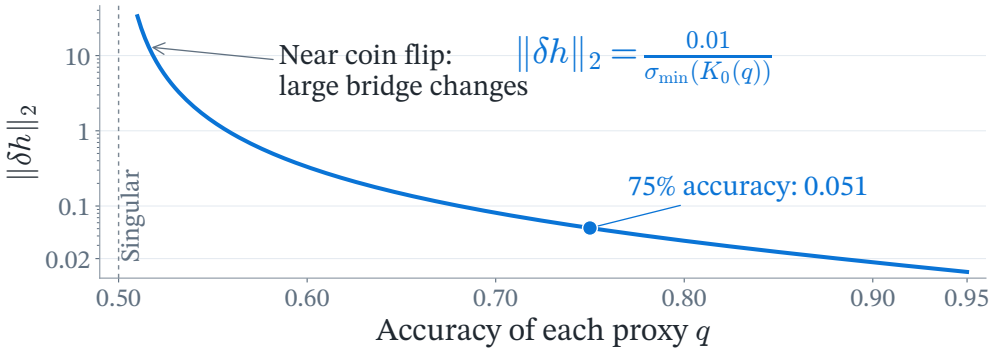
$$\widehat{h}_{t,\lambda} = \arg\min_h \{\|\widehat{K}_t h - \widehat{m}_t\|_2^2 + \lambda \|h\|_2^2\}.$$

3. Average

$\widehat{\mu}(t) = \widehat{p}_W^\top \widehat{h}_{t,\lambda}$.

Finite-dimensional illustration. The numerical and causal questions remain distinct.

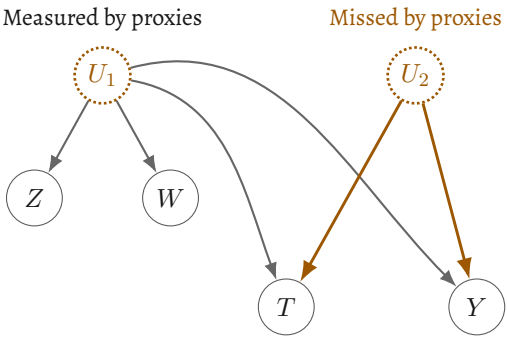
Weak proxies and unstable inversion



Small moment errors can become large bridge errors.

Controlled perturbation with fixed K_t. Vertical axis measures bridge error, not ATE error.

Two proxies can miss a cause



True ATE

0

Observed bridge contrast

$5/8$

The observed system is invertible. Latent completeness fails.

Constructed counterexample, exact enumeration. Dotted nodes are hidden. Full calculation: appendix A7–A8.

Operations and causal assumptions

	Main operation	Key causal link
Adjustment	Average within-C responses	Exchangeability given C
Linear IV	Divide covariances	Common slope, exclusion, exogeneity
Proxy bridge	Solve, then average	Latent exchangeability, restrictions, existence and completeness

A computable formula becomes causal through its identifying assumptions.

Targets and complete qualifications: appendix A15. Numerical success alone does not establish causal validity.

The separate 90-minute practical

15 min	Standardize kidney-stone outcomes
10 min	Compare observationally equivalent models
20 min	Compute IV and violate exclusion
25 min	Solve and average two proxy bridges
13 min	Separate instability from a missing cause
7 min	Debrief

Before calculating: state the target and its causal assumption.

Reading: foundations and instruments

- Hernán and Robins (2020).
  Causal Inference: What If.
- Charig et al. (1986).
  The kidney-stone treatment comparison.
- Imbens (2014).
  Instrumental Variables: An Econometrician's Perspective.
- Angrist, Imbens and Rubin (1996).
  Identification of causal effects using IV.

Reading: proxy identification

- Miao, Geng and Tchetgen Tchetgen (2018).
  Identifying causal effects with proxy variables.
- Tchetgen Tchetgen et al. (2020).
  An Introduction to Proximal Causal Learning.
- Zhang et al. (2023).
  Proximal causal inference without uniqueness.
- Mastouri et al. (2021).
  Proximal causal learning with kernels.

DAG notation and the choice of controls

In a directed acyclic graph, an arrow represents a possible direct causal effect. Missing arrows encode assumptions.

Pattern	Implication for a total treatment effect
$T \leftarrow C \rightarrow Y$	A common cause can create association. Conditioning on a sufficient set of such variables can justify adjustment.
$T \rightarrow M \rightarrow Y$	A mediator carries part of the total effect. Controlling for it changes the question and needs a separate identification argument.
$T \rightarrow S \leftarrow V \rightarrow Y$	Conditioning on the collider S can open an otherwise blocked path and create bias.

These patterns explain why predictive accuracy alone cannot select adjustment covariates.

Pearl (2009). Hernán and Robins (2020), Chapters 6–9.

Outcome regression, weighting and double robustness

For binary treatment, let $m_t(c) = \mathbb{E}[Y \mid T = t, C = c]$ and $e_t(c) = \Pr(T = t \mid C = c)$.

Under the adjustment assumptions, an augmented estimator has form

$$\widehat{\mu}_t = \frac{1}{n} \sum_{i=1}^{n} \left[\widehat{m}_t(C_i) + \frac{\mathbf{1}\{T_i = t\}}{\widehat{e}_t(C_i)} \{Y_i - \widehat{m}_t(C_i)\} \right].$$

With appropriate sampling, overlap and estimation conditions, it is consistent if either the outcome regression or the propensity model is correct. Both may also be correct.

Flexible learned models require suitable sample splitting or empirical-process conditions. Inference needs additional rate and regularity conditions.

Double robustness concerns nuisance estimation. It does not restore exchangeability when the chosen C leaves confounding uncontrolled.

Hernán and Robins (2020), Chapter 13. Formula shown for a simple point-treatment setting.

Binary IV with heterogeneous effects: a different target

Let $Z, T \in \{0, 1\}$ and $T(z)$ be treatment receipt under instrument level z. Assume consistency, no interference and:

- $Z \perp \{Y(0), Y(1), T(0), T(1)\}$ and $0 < \Pr(Z = 1) < 1$.
- Exclusion: $Y(t, z) = Y(t)$.
- Monotonicity: $T(1) \geq T(0)$ almost surely.
- A positive proportion of compliers: $\Pr(T(1) > T(0)) > 0$.

Then

$$\frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[T \mid Z = 1] - \mathbb{E}[T \mid Z = 0]} = \mathbb{E}[Y(1) - Y(0) \mid T(1) > T(0)].$$

This target averages effects among units whose treatment changes with Z. It need not equal the population ATE.

Angrist, Imbens and Rubin (1996). Finite means assumed.

The two proxy restrictions produce the latent residual

Let $a_t(U) = \mathbb{E}[Y \mid T = t, U]$ and $r_t(U) = \mathbb{E}[h_t(W) \mid U]$.

$$\begin{aligned}\mathbb{E}[Y \mid Z, T = t] &= \mathbb{E}\{\mathbb{E}[Y \mid U, Z, T = t] \mid Z, T = t\} \\ &= \mathbb{E}[a_t(U) \mid Z, T = t], \\ \mathbb{E}[h_t(W) \mid Z, T = t] &= \mathbb{E}\{\mathbb{E}[h_t(W) \mid U, Z, T = t] \mid Z, T = t\} \\ &= \mathbb{E}[r_t(U) \mid Z, T = t].\end{aligned}$$

The first equality pair uses $Y \perp Z \mid (T, U)$. The second uses $W \perp (T, Z) \mid U$.

The observed bridge equates the left sides. Latent completeness then forces $a_t(U) - r_t(U) = 0$.

A finite latent model explains the rank condition

Suppose U, Z, W each have exactly k categories, with positive relevant cells. Define

$$Q_t[z,u] = \Pr(U = u \mid Z = z, T = t), \qquad R[u,w] = \Pr(W = w \mid U = u).$$

The proxy restrictions imply

$$K_t = Q_t R, \qquad m_t = Q_t a_t, \qquad p_W^\top = p_U^\top R.$$

If the square K_t is nonsingular, both Q_t and R are nonsingular. Therefore latent completeness holds and a bridge exists:

$$h_t = R^{-1} a_t, \qquad p_W^\top h_t = p_U^\top a_t = \mu(t).$$

The claim depends on the latent cardinality assumption. An observed 2×2 matrix can be nonsingular while missing part of a four-state U.

Miao, Geng and Tchetgen Tchetgen (2018), categorical identification argument.

The causal average can be unique when the bridge is not

Suppose h_t and h'_t solve the same observed equation. Set $d = h_t - h'_t$.

$$K_t d = 0.$$

Under the proxy restrictions and latent completeness,

$$\mathbb{E}\{\mathbb{E}[d(W) \mid U] \mid Z, T = t\} = 0, \qquad \mathbb{E}[d(W) \mid U] = 0.$$

Thus $\mathbb{E}[d(W)] = 0$ and both bridges have the same population average.

In finite linear algebra, the average is constant over the solution set exactly when

$$p_W^\top d = 0 \quad \text{for every } d \in \text{null}(K_t).$$

This algebraic condition concerns uniqueness of the average. Its causal meaning still requires the identification assumptions.

Zhang, Li, Miao and Tchetgen Tchetgen (2023).

An invertible observed matrix can miss a confounding component

Let U_1, U_2 be independent fair bits. Given $U = (U_1, U_2)$:

- Z, W are independent 90%-accurate measurements of U_1.
- Treatment is independent of the proxy noises, with

$$\Pr(T = 1 \mid U) = 0.1 + 0.2U_1 + 0.6U_2.$$

Set $Y(t) = U_1 + U_2$ for both treatment levels. The true ATE is zero.

Consistency, latent exchangeability, positivity and the proxy restrictions hold. Both observed proxy matrices are nonsingular.

Latent completeness fails. The two binary proxies do not resolve the extra confounding contribution of U_2.

Constructed counterexample, evaluated by exact enumeration in the practical. It shows why observed rank alone is insufficient for the causal formula.

The wrong causal answer persists at exact population quantities

For the preceding model, exact enumeration gives

$$K_0 = \begin{pmatrix} 49/58 & 9/58 \\ 3/14 & 11/14 \end{pmatrix}, \quad m_0 = \begin{pmatrix} 9/29 \\ 1 \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 11/14 & 3/14 \\ 9/58 & 49/58 \end{pmatrix}, \quad m_1 = \begin{pmatrix} 1 \\ 49/29 \end{pmatrix}.$$

Both determinants equal $128/203$. Their unique bridges are

$$h_0 = \begin{pmatrix} 9/64 \\ 79/64 \end{pmatrix}, \qquad h_1 = \begin{pmatrix} 49/64 \\ 119/64 \end{pmatrix}, \qquad p_W = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}.$$

Consequently,

$$p_W^\top (h_1 - h_0) = 5/8 \neq 0 = \tau.$$

More observations or a better numerical solver cannot repair this false causal interpretation.

The continuous mean bridge is an integral equation

If $W \mid Z, T = t$ has a conditional density,

$$m_t(z) = \int h_t(w) p(w \mid z, T = t)\, dw = (K_t h_t)(z).$$

This is a Fredholm equation of the first kind. Choose function spaces, for example appropriate L^2 spaces, before discussing its inverse.

- Existence requires m_t to lie in the range of K_t.
- Observed completeness makes K_t injective in the chosen class.
- An injective operator can still have an unbounded inverse.

Sieve, kernel or neural methods need a specified conditional-moment criterion, complexity control, regularization and uncertainty analysis. A small ordinary outcome-prediction error alone does not establish a correct bridge.

Miao et al. (2018). The causal link still uses the separate assumptions of the main theorem.

The same observations can imply different causal effects

Let $E \sim \mathrm{Bernoulli}(0.2)$ denote bit-flip noise.

Model A: treatment causes outcome

$T \sim \mathrm{Bernoulli}(0.5)$, independent of E.

$Y(t) = t \oplus E$, where $\oplus$ is XOR.

$\tau = 0.8 - 0.2 = 0.6$.

Model B: a hidden common cause

$U \sim \mathrm{Bernoulli}(0.5)$, independent of E.

$T = U \oplus E, Y(t) = U$.

$\tau = 0$.

Both models give the same observed law:

$$\Pr(T = 1) = 0.5, \quad \Pr(Y = 1 \mid T = 1) = 0.8, \quad \Pr(Y = 1 \mid T = 0) = 0.2.$$

The observed law of (T, Y) does not generally identify the ATE in this unrestricted causal model.

Other outcome functions require their own bridges

For an outcome transformation f, define $\theta_f(t) = \mathbb{E}[f(Y(t))]$.

If the corresponding integrable bridge exists and the identification conditions hold,

$$\mathbb{E}[f(Y) \mid T = t, Z] = \mathbb{E}[h_{f,t}(W) \mid T = t, Z], \qquad \theta_f(t) = \mathbb{E}[h_{f,t}(W)].$$

Function	Target
$f(y) = y$	Intervention mean
$f(y) = \mathbf{1}\{y \le c\}$	Intervention CDF at c
$f(y) = k(y, v)$	A kernel feature at v

A mean bridge alone does not identify a whole distribution. Existence and regularity must hold for the requested family of functions.

A conditional-moment objective for a continuous bridge

For fixed binary treatment t, define

$$(K_t h)(z) = \mathbb{E}[h(W) \mid Z = z, T = t], \qquad m_t(z) = \mathbb{E}[Y \mid Z = z, T = t].$$

If the required second moments exist,

$$\begin{aligned}&\mathbb{E}\big[(Y - (K_t h)(Z))^2 \mid T = t\big] \\ &\quad = \mathbb{E}[\mathrm{Var}(Y \mid Z, T = t) \mid T = t] + \mathbb{E}\big[(m_t(Z) - (K_t h)(Z))^2 \mid T = t\big].\end{aligned}$$

The first term does not depend on h. The second measures bridge mismatch.

In practice, estimate K_t, choose a function class, regularize and assess uncertainty. Flexible nuisance fits require appropriate sample splitting or complexity control.

Mastouri et al. (2021); Gretton, MBZUAI 2026 Part 2, slide 33. The objective alone does not establish causal identification or stable estimation.

Sample 2SLS: full matrix statement

Center $\mathbf{y}, \mathbf{t}$ and the instrument columns $\mathbf{Z}$. Require full column rank of $\mathbf{Z}$ and $\mathbf{t}^\top P_{\mathbf{Z}} \mathbf{t} > 0$.

$$P_{\mathbf{Z}} = \mathbf{Z}(\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top, \qquad \widehat{\mathbf{t}} = P_{\mathbf{Z}} \mathbf{t}.$$

$$\widehat{\beta}_{\text{2SLS}} = \frac{\widehat{\mathbf{t}}^\top \mathbf{y}}{\widehat{\mathbf{t}}^\top \widehat{\mathbf{t}}} = (\mathbf{t}^\top P_{\mathbf{Z}} \mathbf{t})^{-1} \mathbf{t}^\top P_{\mathbf{Z}} \mathbf{y}.$$

The first stage is a sample linear projection. Intercepts may equivalently be included in both stages. Use IV standard errors.

Imbens (2014). This estimates the common slope under the model and restrictions in the main lecture.

Proxy mean identification: sufficient conditions

For each binary treatment level t, assume:

1. Well-defined treatments, no interference, consistency, and $Y(t) \perp T \mid U$, $\Pr(T = t \mid U) > 0$ almost surely.
2. Proxy restrictions: $Y \perp Z \mid (T, U)$ and $W \perp (T, Z) \mid U$.
3. An integrable h_t solves $\mathbb{E}[Y \mid Z, T = t] = \mathbb{E}[h_t(W) \mid Z, T = t]$.
4. For every $g \in L^1(P_U)$, $\mathbb{E}[g(U) \mid Z, T = t] = 0$ almost surely implies $g(U) = 0$ almost surely.

Require $\mathbb{E}|Y(t)| < \infty$ and $\mathbb{E}|h_t(W)| < \infty$. Then

$$\mu(t) = \mathbb{E}[h_t(W)].$$

Observed completeness for functions of W concerns a different question: uniqueness of the bridge, conditional on existence.

Miao et al. (2018); Tchetgen Tchetgen et al. (2020). One sufficient route, with latent completeness stated explicitly.

Method comparison: targets and failure modes

	Adjustment	Linear IV	Proxy bridge
Observed	(C, T, Y)	(Z, T, Y)	(Z, W, T, Y)
Causal link	Exchangeability given C	Common slope, exclusion, exogeneity	Latent exchangeability, restrictions, existence and latent completeness
Target here	$\mu(t)$	β	$\mu(t)$
Difficulty	Residual confounding, overlap	Invalid or weak instrument	Invalid proxies, omitted latent component, inverse sensitivity

Maintain consistency, suitable support/positivity and the required moment conditions. The tables name key links, not every regularity condition for an estimator.

Hernán and Robins (2020); Imbens (2014); Miao et al. (2018).

The constructed airline population

Let U, Z, V, η be independent standard normal variables.

$$T = 110 + 10Z + 10U + 5V, \qquad Y(t) = 250 - 0.5t + 20U + 5\eta.$$

The exact population relationships are

$$\mathbb{E}[Y \mid T = t] = 195 + \tfrac{7}{18}(t - 110), \qquad \mathbb{E}[Y(t)] = 250 - 0.5t,$$

$$\mathbb{E}[T \mid Z = z] = 110 + 10z, \qquad \mathbb{E}[Y \mid Z = z] = 195 - 5z.$$

Within $Z = z$, the observed slope is 1.1. Across instrument means, it is -0.5.

The plotted $Z = -1, +1$ levels are points of a continuous instrument. The separate Wald table is a binary-instrument illustration.

Constructed linear population, not airline data. The sales outcome is treated as continuous in this illustration.

Both exact binary proxy systems

With row and column order $(0, 1)$,

$$K_0 = \begin{pmatrix} .7 & .3 \\ .5 & .5 \end{pmatrix}, \quad m_0 = \begin{pmatrix} .4 \\ 2 \end{pmatrix}, \qquad K_1 = \begin{pmatrix} .5 & .5 \\ .3 & .7 \end{pmatrix}, \quad m_1 = \begin{pmatrix} 4 \\ 5.6 \end{pmatrix}.$$

Solving $K_t h_t = m_t$ gives

$$h_0 = \begin{pmatrix} -2 \\ 6 \end{pmatrix}, \qquad h_1 = \begin{pmatrix} 0 \\ 8 \end{pmatrix}, \qquad p_W = \begin{pmatrix} .5 \\ .5 \end{pmatrix}.$$

Thus $\mu(0) = 2$, $\mu(1) = 4$ and $\tau = 2$ under the verified generator and causal assumptions.

Exact enumeration. Ordinary adjustment for W gives means 1.2 and 4.8, and contrast 3.6.

Figure sources and provenance

- Procedure illustration: NIDDK media asset 17731. Credit: National Institute of Diabetes and Digestive and Kidney Diseases, National Institutes of Health. Reused under the NIDDK reuse policy.
- Kidney-stone charts: exact reported counts from Charig et al. (1986).
- IV charts: exact analytic relationships from the constructed population in appendix A16.
- Proxy charts: exact enumeration of the binary generator; the stability chart states its fixed-operator perturbation.

Native diagram sources and editable vector charts accompany the deck.